Worksheet

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24 February 2025

1 Gravitation Law - Practice Problem

(a)What is the force of attraction between the Sun and the Earth?

Values needed: Mass of the Sun $M_S = 2 \times 10^{30}$ kgs. Mass of the Sun $M_E = 6 \times 10^{24}$ kgs. Distance between the Sun and the Earth is 149 million km.

(b) If Superman is standing in Columbus, what is the force of gravity on him?

Values needed: Mass of Superman: 100 kg Radius of Earth: 6400 km

(c) What is the force of attraction between the Sun and Superman? (We can assume that the distance between the Sun and the Earth is the same as the distance between the Sun and Superman when he is standing in Columbus.)

2 Ideal Gas Law - Brief Review

(a) The ideas gas law is $PV = nRT$, where n is number of moles. If we want to write this as $PV = NkT$, where N is the number of molecules and k is a new constant, then what should be the value of k?

(This k is the Boltzmann constant. Check the value on the internet and see if your calculation is right)

(b) If the internal energy of the ideal gas is written as $U = \frac{3}{2}NkT$, then find the pressure of the gas in terms of internal energy and volume.

3 Log-Log plots

(a) Plot these following curves in your choice of plotting tool.
(i) $y = x$ (ii) $y = x^2$ (iii) $y = x^7$

(b) Now, we will plot the same curves in the log-log scale. So, instead of each point of the plot being (x, y), it would be $(\log x, \log y)$.

4 Mass and Luminosity of stars

(a) Take your favourite five stars (make sure they are not the same kind of stars) and list their mass and temperature and luminosity.

(b) Make a plot of their luminosity vs their masses.

(c) Make a plot of their luminosity vs their temperatures.

(d) Now, make log-log plots for the same curves.

(This problem is a little hard, and open-ended, and there probably will be issues (maybe with units etc) to solve it, but please don't worry about it. Try to do as much as you can, and we will talk about this in the next class. This is aimed to give you a flavour of how "independent" research or problem-solving looks like. Have fun with it!!)